SUPPORTING ADULT GUIDANCE

Algebraic proof

Algebraic Expressions and Identities · Lesson 7. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson makes proof the main aim. Learners first choose algebra for kinds of number, order the lines of two proofs, and judge arguments as proofs, mere examples or flawed proofs. In the main activity, the conjecture lab, each claim is tested with the learner's own numbers, judged always, sometimes or never true, and then either disproved with a counterexample or proved: choosing a representation, writing each algebra line (checked by value and by form), and choosing the conclusion.

Driving Question: *How can I be sure it is true for every number?*

The mathematics, in plain terms

An algebraic proof represents arbitrary integers symbolically ($2n$, $2n + 1$, $n + k$) and derives the required form by valid manipulation, so the result holds for every integer. A single counterexample disproves a universal claim. Independent quantities need independent variables: using $2n + 1$ for both of two odd numbers proves the claim only for squares.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Test, decide, prove

Claim: the sum of any three consecutive whole numbers is a multiple of 3.

1. Test: 4 + 5 + 6 = 15, 10 + 11 + 12 = 33.	both multiples of 3
2. Write the numbers in algebra.	n, $n + 1$, $n + 2$
3. Add and simplify.	$n + (n + 1) + (n + 2)$ $= 3n + 3$
4. Factorise.	$= 3(n + 1)$
5. Conclude.	3 × a whole number, so always a multiple of 3

Proved: the sum of any three consecutive whole numbers is a multiple of 3. ■

Example 2 — A counterexample

Claim: $n^2 + n + 41$ is always prime.

1. Test n = 0, 1, 2, 3: 41, 43, 47, 53.	all prime
2. It stays prime all the way to n = 39.	forty examples
3. Test n = 40: 1600 + 40 + 41 = 1681.	= 41 × 41
4. 1681 is not prime, so the claim is false.	one counterexample

Not always prime — however many examples work, one counterexample is enough.

Evidence, disproof, proof

Lesson 1 showed that testing can suggest and a counterexample can disprove; this lesson completes the picture with proof. The jigsaw shows the shape of a proof before learners write one, the critic trains judgement, and the lab puts it all together. The $n^2 + n + 41$ claim is deliberately included: forty successes, then a failure.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Write it in algebra: even, odd, consecutive, multiples.	~6 min
Movement	The proof jigsaw: order the lines of two proofs.	~6 min
Reflection	The proof critic: proof, only examples, or a flawed step?	~7 min
Creativity	The conjecture lab: test, decide, then prove or disprove. Stretch: why two letters.	~21 min
Rest	A claim that held forty times and then broke.	~2 min
Nutrition	Intellectual: the power of a good argument.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Believing several examples prove a claim.
Ask: “Have you checked every number? Could the next one break it?”

Using n + 1 for an odd number.
Ask: “What is n + 1 when n = 2?”

Using the same letter for two different numbers.
Ask: “If both odd numbers are $2n$ + 1, are they different numbers?”

Stopping at $3n$ + 3 without explaining why it is a multiple of 3.
Ask: “Can you write it as 3 × something?”

Thinking a counterexample needs many examples.
Ask: “If a claim says ‘always’, how many failures does it take to break it?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: odd and even numbers, multiples, and expanding and factorising. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner proves 'the product of two odd numbers is odd' using $(2n + 1)(2n + 1)$. The aim is a question that lets them see the gap.

- “Which two odd numbers does your proof cover when n = 2?”
- “Does it cover 3 × 5?”

- “How could you let the two numbers be different?”
- “What changes in the algebra?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Are the conclusions required in words?
Yes: a proof is not finished until it says what has been shown. The lab asks for it every time.

Why include a claim that is false only at $n = 40$?
It shows vividly that no number of examples is a proof — the heart of the lesson.

Is the stretch required?
No. The secure core is the first three proofs and the counterexample.

What comes next?
Lesson 8: function notation, domain and range — the final lesson of the unit.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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